

NIRC2-Pol: First Light of Near-Infrared Polarimetry on Keck II

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ABSTRACT

NIRC2, the Near Infrared Camera 2 on the Keck II telescope, was recently upgraded with a new suite of polarimetric observing modes. The new polarimetry modes (referred to as NIRC2-Pol) open up a wide range of new studies, including investigations of exoplanets, the Galactic center, active galactic nuclei and the solar system. The new modes enabled by the upgrade span the 1.1 to 4.1 micron range (i.e. J through L' bands) and include imaging polarimetry, coronagraphic imaging polarimetry, and low resolution spectropolarimetry. NIRC2-Pol is unique, as Keck II is the largest telescope (10 m) on which infrared polarimetry capabilities are available, one of few with L' polarimetric imaging, and the only on which there is both a polarimetric mode and a vortex coronagraph. Here, we introduce the design of NIRC2-Pol, its capabilities, its current operational status, and its first on-sky results: the first L' polarimetric images of the AB Aurigae circumstellar disk. These images more clearly reveal the disk's iconic spiral arms than previous L' imaging, and no brightness enhancement is seen at the location of the candidate protoplanet AB Aur b, which would be expected in polarized light if the feature were a compact disk structure such as a dust clump.

Keywords: Polarimetry; Infrared Astronomy; Instrumentation

1. INTRODUCTION

Polarimetry is a useful technique for studying various astronomical scenes, including objects with strong magnetic fields or where there is light scattering off dust grains. It is particularly useful for high-contrast imaging of circumstellar disks, as it enables polarimetric differential imaging (PDI); light from a disk is polarized due to scattering, whereas starlight is generally unpolarized, creating a natural separation between the two and therefore a way to remove starlight without disk self-subtraction (Follette 2023). However, polarimetry is still

a relatively rare capability on large telescopes, especially when combined with high-contrast imaging capabilities and/or at thermal infrared wavelengths such as L' band. Polarimetric capabilities in the infrared (IR) currently exist with VLT/SPHERE/IRDIS (De Boer et al. 2020; van Holstein et al. 2020), Subaru/SCEXAO/CHARIS (GJ't Hart et al. 2021), and Gemini/GPI (Perrin et al. 2010; Millar-Blanchaer et al. 2016; Chilcote et al. 2020); however, these three instruments only operate up to K band ($\lambda_c \sim 2.2\mu\text{m}$), excluding thermal IR wavelengths such as L' band ($\lambda_c \sim 3.8\mu\text{m}$), where only Subaru/IRCS is available (Terada et al. 2018; Watanabe et al. 2018). These instruments have used polarimetry to produce many results of interest, such as: spectropolarimetry of the transition disk HD 34700 A (Subaru/SCEXAO/CHARIS, Chen et al.

(2024)); polarimetric imaging of brown dwarfs, embedded protoplanets, and disks around evolved binaries (VLT/SPHERE/IRDIS, van Holstein et al. (2021), Wahhaj et al. (2024), and Andrych et al. (2023)); and deep characterization of the HR 4796A debris disk in polarized light (Gemini/GPI, Perrin et al. (2015) and Arriaga et al. (2020)). Beyond planetary and stellar astronomy, polarimetry is also a technique of interest for the galactic center and active galactic nuclei, e.g. Nishiyama et al. (2008); Blinov et al. (2021). A number of past instruments were also capable of polarimetric imaging, but have since been decommissioned, such as Subaru/HiCIAO (Hodapp et al. 2006; Tanii et al. 2012) and VLT/NACO (Witzel et al. 2011; de Regt et al. 2024).

In this work, we present a key addition to the current roster of imaging polarimeters on major telescopes: NIRC2 Polarimetry (NIRC2-Pol). NIRC2, the Near Infrared Camera 2 for Keck Observatory, is a near-infrared imager operating from ~ 1 to $5 \mu\text{m}$ (Y through M bands) on the 10 meter Keck II telescope on Mauna Kea, positioned behind the telescope’s facility adaptive optics system (van Dam et al. 2004; Wizinowich et al. 2006; Lilley et al. 2024). The instrument is kept in a cryogenic dewar, and the optical path contains two filter wheels with pupil masks and both narrowband and broadband filters, plus grisms, slits for spectroscopy, and coronagraphic spots on slide mechanisms. In addition to traditional coronagraphs, NIRC2 has two vector vortex coronagraph masks, one optimized for L' and M bands and another for K band (Castellá et al. 2016; Xuan et al. 2018). NIRC2 is used for a variety of science cases including exoplanet imaging, tidal evolution near Sgr A* (Ciurlo et al. 2023), planet formation around young stars (Uyama et al. 2020), and more.

NIRC2 has now been upgraded to enable a new suite of polarimetric observing modes, including imaging polarimetry, high-contrast coronagraphic imaging polarimetry, and low resolution spectropolarimetry. To create polarimetric capabilities in NIRC2, two additional hardware components were added: a Wollaston prism to split the beam into two polarization states, and a half-waveplate to modulate the angle of polarization for calibration. This upgrade began in 2019 with the installation of the Wollaston prism in one of NIRC2’s filter wheels, and was recently completed with the installation of two half-wave plate modulators (one for $J/H/K$ and one for L') in August 2025. Commissioning observations were completed during Semester 25B, and the mode recently became available to the community starting in Semester 26B.

NIRC2-Pol is unique, as Keck II is the largest telescope (10 m) on which IR polarimetry capabilities are available. It has a broad wavelength range—spanning J through L' , ~ 1.1 to $4 \mu\text{m}$ —where polarimetry in L' is a particularly unique feature. NIRC2-Pol has access to the multitude of other features already available on NIRC2 and Keck II adaptive optics (AO), including grisms for spectropolarimetry, many options of coronagraphic masks, adaptive optics using both a natural guide star (NGS AO) or a laser guide star (LGS AO), and more. NIRC2-Pol is notably the only polarimeter equipped with a vortex coronagraph, making this an interesting testbed for such observations which may be of substantial interest to upcoming missions in development such as the *Habitable Worlds Observatory* (Feinberg et al. 2024). Vortex coronagraphs generally have a superior inner working angle (Mawet et al. 2009; Haffert et al. 2025), and polarimetry enables a number of interesting investigations into exoplanet characterization (Vaughan et al. 2023; Gordon et al. 2025). NIRC2-Pol’s role in the current landscape of astronomical polarimeters on large telescopes is summarized in Table 1.

In this letter, we briefly introduce the design of the NIRC2 Polarimetry mode, its capabilities, and its first on-sky results, with the goal of sharing this new mode with the community and encouraging its use for a variety of science cases.

2. NIRC2 POLARIMETRY MODE DESIGN

NIRC2-Pol is a dual-channel polarimeter, meaning that it uses a polarizing beamsplitter to split the incoming light into two orthogonal polarization states and a half-wave plate (HWP) to modulate the angle of polarization. Through cycles of four critical HWP angles (0° , 45° , 22.5° , 67.5°), it is possible to recover the Stokes vector components I , Q and U (Stokes 1851; De Boer et al. 2020) from the two orthogonal polarization states recorded on the detector. These components can be used to measure the degree and angle of linear polarization.

Accordingly, NIRC2-Pol relies on two additional optics in the Keck II AO bench and NIRC2—a HWP and a Wollaston prism—plus a $5\times 10''$ field mask in NIRC2’s slit and mask sliders to reduce the FOV such that the ordinary and extraordinary beams do not overlap (see Fig. 1). The Wollaston prism, a polarizing beamsplitter, is located in the NIRC2 outer filter wheel, and the HWP is located in the Precision Calibration Unit (PCU2), which is at the front (telescope side) of the Keck II AO bench.

This mode was designed as an upgrade to the existing NIRC2 instrument, as part of a new Precision Calibration Unit (PCU2) analogous to the Pinhole Calibration Unit (PCU1) on Keck I (Freeman et al. 2023; Claveau

Instrument	LGS AO	IR WFS	Wavelengths Covered				Coronagraphs	Telescope Size
			J	H	K	L'		
VLT/SPHERE/IRDIS	✓	✓	✓	✓	✓		✓	8.2 m
Subaru/SCEXAO/CHARIS	✓	✓	✓	✓	✓		✓	8.2 m
Gemini/GPI	✓	✓ [†]	✓	✓	✓		✓	8.1 m
Subaru/IRCS	✓	✓	✓	✓	✓	✓	✓	8.2 m
Keck/NIRC2-Pol	✓	*	✓	✓	✓	✓	✓ [‡]	10 m

Table 1. Various capabilities for infrared polarimeters available on 8-10 meter class telescopes. NIRC2-Pol is one of only two polarimeters capable of operating in L' band with LGS AO and coronagraphy (alongside IRCS (Terada et al. 2018)), and is the the only such polarimeter on a 10 meter telescope. [†]IR WFS capabilities are planned as part of the ongoing GPI 2.0 upgrade and move to Gemini North (Chilcote et al. 2020). *IR WFS capabilities are in progress as part of an AO upgrade project at WMKO (Lilley et al. 2024). [‡]NIRC2-Pol is the only polarimeter equipped with a *vortex* coronagraph.

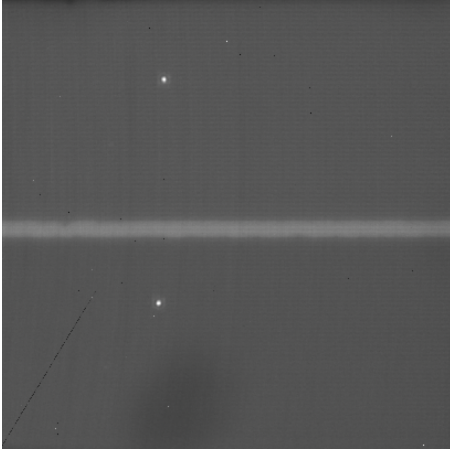


Figure 1. A typical image of a star with the NIRC2-Pol optics—a half-wave plate, a Wollaston prism polarizing beamsplitter, and a 5x10" mask—in the beam. Defects typical of NIRC2 imagery (e.g. the diagonal line, the thumbprint, striping on the detector) are present, along with a bright band in the center of the image. This band is where the fields from the ordinary and extraordinary beams overlap, as the same field is being imaged twice on the same detector in two orthogonal polarization states, reducing the field-of-view from 10x10" to 5x10".

et al. 2026). The PCU2 contains the optomechanics necessary to rotate and move the HWPs or, alternatively, a pinhole mask that can be used to analyze distortion on the detector for precision astrometry (Lin et al. 2024). The PCU2 has a mount in a rotation stage that can hold one optic at a time out of the following options: a pinhole mask, a half-wave plate optimized for $J/H/K$ -band polarimetric observations, or a half-wave plate optimized for L' polarimetric observations. To switch between these three, the optic in the PCU2 must be manually switched by WMKO staff. This swap procedure is a limiting factor for observations, and must be accounted for in proposals and observation planning, as the observatory requires significant notice to plan and perform this task.

2.1. HWP Specifications

The JHK HWP was fabricated by Bernard Halle and is effective over the wavelength range 1.1–2.4 μ m. It is an achromatic half wave retarder made of two plates—one quartz (1.48 mm thickness) and one MgF₂ (1.16 mm thickness)—with an air gap and four spacers. The fast axis is marked with dots on the side surface, and the waveplate has a clear aperture of 53 mm. The quartz side of the JHK HWP is coated with TelAztek RAR Nano-Texture. The throughput of the JHK HWP is over 92% across the wavelength range tested in lab (1.1–1.9 μ m), and is predicted to be this high or better at longer wavelengths, according to theoretical values from the manufacturer. The manufacturer specification for the JHK HWP retardance is $\lambda/2 \pm \lambda/70$, as confirmed in lab measurements. We note that the JHK HWP was damaged in transit to the observatory, and as a result has a small crack on the edge of the plate. This defect is not in the beam path with standard target placement in the center of the field.

The L' waveplate was fabricated by the Karl Lambrecht Corporation and is effective over the wavelength range 3.4–4.1 μ m. It is a zero order double plate air spaced MgF₂ half wave retarder (2.75 mm thickness) with a clear aperture of 53 mm. Its throughput, as specified by the manufacturer, is >90% across its effective wavelength range. To determine the L' plate's retardance, laboratory measurements were taken using a dual-rotating retarder polarimeter (DRRP) between 1-2 microns and extrapolated to L' wavelengths, finding less than $\lambda/5$ variation across L' band (Melby et al. 2024).

3. OBSERVING WITH NIRC2-POL

NIRC2-Pol is a non-facility visitor mode on Keck II as of Semester 26B. A number of tools have been developed to assist an observer with NIRC2-Pol observations, such as observing scripts, an updated calculator for observing overheads, and a GUI to show the HWP

angle, 5x10" mask status, and PCU2 position. The ZetaPersei package provides tools for planning polarimetric standard stars and polarimetric calibration for NIRC2 (Zhang et al. 2025), and a framework for polarimetric signal-to-noise ratio calculations is available in Nguyen et al. (2021). A full operations guide is also available, with detailed instructions for observing with the new mode.¹

3.1. Executing Polarimetric Observations

To recover the Stokes components Q and U (and therefore the degree of linear polarization), polarimetric observations must be executed in cycles of four critical HWP angles: 0° , 45° , 22.5° , and 67.5° . NIRC2-Pol has an observing script that allows the user to specify a number of HWP cycles to execute, with the option for dithering on the target. Generally, a HWP cycle should aim for a total time of $\lesssim 4$ minutes (1 minute per angle) to avoid smearing of any time-dependent changes in the polarization signal, and so integration times and coadds should be chosen accordingly.

There are three modes currently available and tested on-sky with NIRC2-Pol in both JHK and L' : standard imaging polarimetry, high-contrast imaging polarimetry with traditional Lyot coronagraphs, and high-contrast imaging polarimetry with a vortex coronagraph. As previously stated, Keck II is the only major facility with both polarimetric capabilities and a vortex coronagraph. A vortex coronagraph has spatially-varying retardance, therefore altering the polarization vector of incoming light and making data processing more complex. A data processing pipeline for NIRC2-Pol—including an effort to incorporate vortex observations and polarimetry—is currently in development (Lewis et al. in preparation). Low-resolution spectropolarimetry with the NIRC2 grisms is theoretically possible, but has not been tested and will not be included in official commissioning activities.

Observing in L' also presents challenges due to the substantially higher thermal background. Dome flats without the Wollaston in and the full detector size are not generally possible, as they will saturate the detector. The thermal background changes on quite short timescales and significant effort is required to remove this effect, as described in Nguyen et al. (2025). With NIRC2-Pol, we recommend taking sky flats/backgrounds with each target for 3 HWP cycles and dithering during observations with an ABBA pattern whenever possible.

¹ <http://bit.ly/3NfomHa>

Although flat fielding can be done in the standard way (i.e. no polarimetric optics in the beam), it is recommended to take “polarimetric flats” to account for imperfections introduced by the additional optics. These flats should be taken with the Wollaston and HWP in the beam, and the HWP rotating continuously; this is a similar procedure to that done by Subaru/SCEXAO/CHARIS (Lozi et al. 2020) and Subaru/SCEXAO/VAMPIRES (Lucas et al. 2024).

3.2. Polarimetric Efficiency

Observers must consider the angle of the image derotator (IMR, a.k.a. K-mirror) during polarimetric observations, as it strongly affects the instrument’s polarimetric efficiency (i.e. how much linear polarization is converted to circular polarization, therefore making it unmeasurable by this type of polarimeter). This is because the IMR acts as a highly-imperfect waveplate and introduces significant instrumental polarization due to the high angle of incidence of the incoming light onto its mirrors. The polarimetric efficiency of the instrument is most highly affected by the IMR at shorter wavelengths. In J and H bands, the angle of the IMR that produces the best polarimetric efficiency is 45 degrees; angles around 0 and 90 should be avoided when possible, especially for quantitative polarimetry. In K band, the same angles should be considered but the effect is less severe. In L' band, the polarimetric efficiency is generally stable regardless of IMR angle.

3.3. Effects on Adaptive Optics Performance

Keck II Adaptive Optics (AO) can be used with NIRC2-Pol as with standard NIRC2 operations, with minor impacts on the Strehl ratio. From commissioning observations in JHK bands, we measured an average Strehl of $56.6 \pm 0.9\%$ with standard NIRC2 observations and an average Strehl of $52.0 \pm 2.2\%$ with NIRC2-Pol; we note that seeing degraded slightly throughout the course of these observing sequences (from $\sim 0.20''$ to $\sim 0.24''$), with high cirrus clouds intermittently present, which may explain the slight discrepancy between the average Strehls.

Strehl is noticeably worse with polarimetry in L' band. During commissioning, we observed an average Strehl of $40.3 \pm 2.7\%$ with the polarimetric optics, and an average of $47.2 \pm 4.2\%$ without. Removing the HWP did not improve Strehl, indicating that NCPAs from the Wollaston are a likely culprit for the degradation of the correction. Fast and Furious focal plane wavefront sensing (Bos et al. 2021) works without any modifications, and showed $\sim 5\%$ improvement to the Strehl in L' band with NGS AO.

Commissioning tests reveal similar degradation of Strehl in L' band with the polarimetric optics for LGS AO. We observed an average Strehl of $18.8 \pm 4.1\%$ with the polarimetric optics, and an average of $25.8 \pm 5.3\%$ without. The field of regard is approximately $30''$ with the HWP in position, placing constraints on choices of tip/tilt stars. The L' HWP reduces the laser throughput to the WFS camera by approximately ± 0.2 magnitudes, an amount that should not affect LGS performance. LGS AO has not been tested for the *JHK*, although similar performance is expected.

3.4. Polarimetric Calibration

A number of calibrations are necessary for polarimetric observations, especially if any quantitative measurements (e.g. degree of linear polarization) are desired. As the HWP is periodically swapped in and out, we recommend taking a sequence of images with HWP angles 0 to 180 degrees after each installation to confirm the fast axis orientation of the waveplate. The HWP mount is designed such that the installation angle should be consistent and repeatable to high precision, but we complete this step as an extra precaution until we have a better understanding of the stability of the instrument. The fast axis corresponds to the angle at which we find a peak in the observed intensity.

A full Mueller Matrix model of the system is currently in development (Zhang et al. in preparation); this model will allow us to correct for instrumental polarization, enabling high-precision quantitative polarimetry similar to CHARIS and VAMPIRES on Subaru/SCEXAO (GJ't Hart et al. 2021; Zhang et al. 2023). **POL CAL ROUGH ESTIMATES FROM REBECCA GO HERE** Without the Mueller Matrix model, NIRC2-Pol is still capable of relative and qualitative polarimetry, including polarimetric differential imaging for circumstellar disks (Follette 2023). Other calibrations—such as dark subtraction and sky subtraction—are done in the standard manner for NIRC2.

4. FIRST ON-SKY RESULTS

During commissioning, we observed the well-studied pre-transitional disk around AB Aurigae (Figure 2), resolving it for the first time in L' polarimetry. AB Aur was observed with non-coronagraphic polarimetric imaging on 7 Dec 2025 for a total integration time of 35.8 minutes, with individual exposures of 0.45 seconds and 45 coadds, the detector in MCDS ($n_{\text{samp}}=4$) read-out mode, and two dither positions.

4.1. Data Processing

The data were first dark-subtracted and flat-fielded, and bad pixels were cleaned up using median replace-

ment with a 7×7 pixel window. The data were then split into two halves for the ordinary/extraordinary beams, the mean thermal background of each frame was subtracted, and the frames were roughly centered. To retrieve Stokes Q and U , we take double differences, where I_1 and I_2 are the two orthogonal polarization states on the detector for a given HWP angle:

$$Q = \frac{1}{2}(Q^+ - Q^-) = \frac{1}{2}(I_1(0^\circ) - I_2(0^\circ) - I_1(45^\circ) - I_2(45^\circ)) \quad (1)$$

$$U = \frac{1}{2}(U^+ - U^-) = \frac{1}{2}(I_1(22.5^\circ) - I_2(22.5^\circ) - I_1(67.5^\circ) - I_2(67.5^\circ)) \quad (2)$$

We then rotate these polarizations from the instrument frame of reference to the sky frame, using a rotation matrix that accounts for the sky position, image rotator position relative to the bench, and the offset of the HWP fast axis from zero (in this case, 59.7° as measured from daytime calibrations); in terms of the relevant NIRC2 header keywords, this $\theta_{\text{rot}} = \text{-PARANG} + \text{EL} + \text{ROTPDEST} + 59.7^\circ$. After this double-differencing procedure, we finally de-rotate the frames (as these observations were taken in NIRC2's **VERTANG** mode) using `pyklip.rotate` (Wang et al. 2015; Wang et al. 2015). The position angle of each frame is calculated as the parallactic angle (**PARANG**) plus the image rotator position relative to the sky (**ROTPOSN**) minus the NIRC2 instrument offset angle (**INSTANGL**) plus an offset as reported in Service et al. (2016).

We then transform to Q_ϕ and U_ϕ radial Stokes parameters, where ϕ is the azimuthal angle around the disk; in this frame, Q_ϕ contains disk signal and U_ϕ should contain noise:

$$Q_\phi = Q \cos(2\theta) - U \sin(2\theta) \quad (3)$$

$$U_\phi = Q \sin(2\theta) + U \cos(2\theta) \quad (4)$$

Double-differencing in this manner removes instrumental polarization downstream of the HWP; however, without a full Mueller matrix model of the system, there is likely still instrumental polarization from M3 and crosstalk between polarization channels. To further remove the residual speckle pattern and roughly remove instrumental polarization, we perform a minimization of U_ϕ on each frame to find the best fit center positions and two instrumental polarization terms (one each for Q and U). To create the final combined image, we perform a weighted combination where the frames are weighted inversely to the noise (i.e. those with the lowest standard deviation in U_ϕ —the least noise—are weighted the highest).

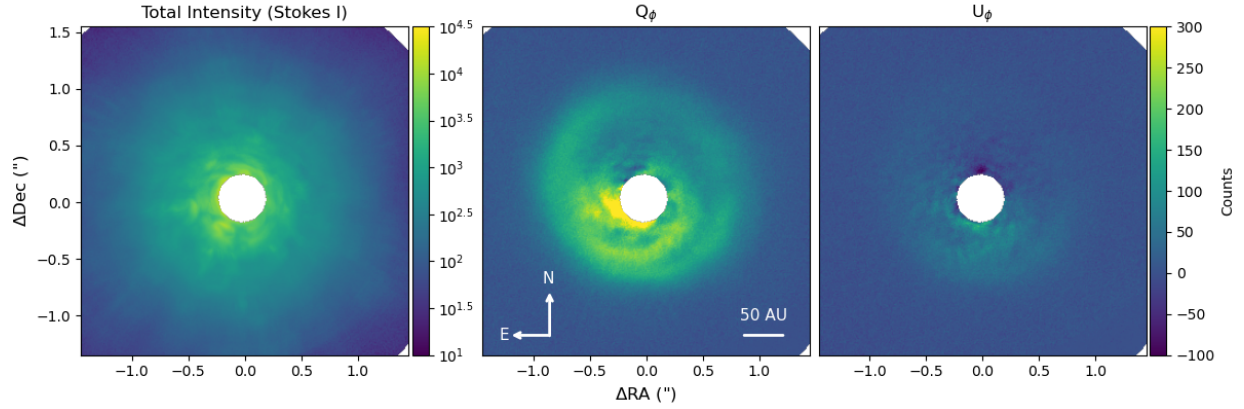


Figure 2. The AB Aur circumstellar disk as seen by NIRC2-Pol during commissioning in L' total intensity (left), Stokes Q_ϕ (middle), and Stokes U_ϕ (right).

Total intensity can be recovered from polarimetric observations by averaging the double sums (I_Q and I_U) derived from the $0/45^\circ$ and $22.5/67.5^\circ$ cycles respectively. For comparison, the recovered total intensity polarimetric data was also reduced as a typical angular differential imaging dataset (see Figure 3). The two polarization components were sub-pixel centroid aligned prior to adding them together. The full aligned data cube was reduced using principal component analysis (PCA) with temporal filtering to reduce self-subtraction. The highest SNR data was produced by subtracting 15 PCA components rejecting frames that had not rotated by at least 40 pixels at the innermost separation. The rotation over the night totaled 74° , meaning each frame had several reference frames for PCA and subtraction.

4.2. Notable Features of AB Aur

AB Aurigae is one of the nearest ($d=162.9\pm1.5$ pc, Brown et al. (2018)) and brightest Herbig Ae stars ($\sim 2-4$ Myr old), and it hosts a well-studied transitional disk with spiral arms and a potential embedded protoplanet (Boccaletti et al. 2020; Currie et al. 2022; Bowler et al. 2025). Spiral arms are generated during the planet formation process as a protoplanet interacts with the disk (Bae & Zhu 2018). The AB Aur disk has been imaged in scattered light at multiple optical and NIR wavelengths, including with polarimetry, but this work presents the first L' polarimetric imaging of the system, shown in Figure 2. With polarimetric differential imaging, we obtain a factor of $\sim 10^2$ in speckle suppression without the use of a coronagraph.

In these L' polarimetric images of the system, we clearly recover the S1 and S2 inner spirals as identified in H -band scattered-light imaging from Boccaletti et al. (2020) (Figure 3), also previously detected in infrared imaging from HiCIAO (Hashimoto et al. 2011) and ALMA CO imaging (Tang et al. 2017). The new

NIRC2-Pol images, thanks to polarimetric differential imaging, reveal significantly more structure than previous L' imaging with the Large Binocular Telescope (LBT), which were poorly resolved due to limitations of the angular differential imaging (ADI) and reference differential imaging (RDI) techniques employed (Jorquera et al. 2022; Betti et al. 2022).

The protoplanet candidate (AB Aur b), however, is distinctly missing from the new L' polarimetry images. First detected in 2022 via Subaru/SCEXAO/CHARIS and *Hubble* imaging (Currie et al. 2022), AB Aur b has been highly debated. AB Aur b was not seen in L' observations with LBT (Jorquera et al. 2022), nor was it detected in Paschen β imaging of the system (Biddle et al. 2024). Zhou et al. (2023) found that the UV-optical emission of the system was inconsistent with that expected for an accreting protoplanet, while $H\alpha$ observations with *Hubble*/WFC3 were inconclusive (Zhou et al. 2022; Bowler et al. 2025) and those from VLT/MUSE were interpreted as evidence of accretion onto AB Aur b (Currie et al. 2025). The lack of detection in the images presented here does not indicate the absence of a protoplanet. Instead, the lack of higher polarized intensity at the location of the candidate serves as mild evidence *against* the theory that the bright spot is a compact disk feature, such as a dust clump; we would expect that a high concentration of dust would appear brighter in polarized intensity. A more thorough investigation of the contrast and planet detection limits in these images is beyond the scope of this brief letter, and is left for future work.

5. CONCLUSIONS

NIRC2 Polarimetry (NIRC2-Pol) enables infrared polarimetric imaging and low-resolution spectropolarimetry on Keck II across J through L' bands. NIRC2-

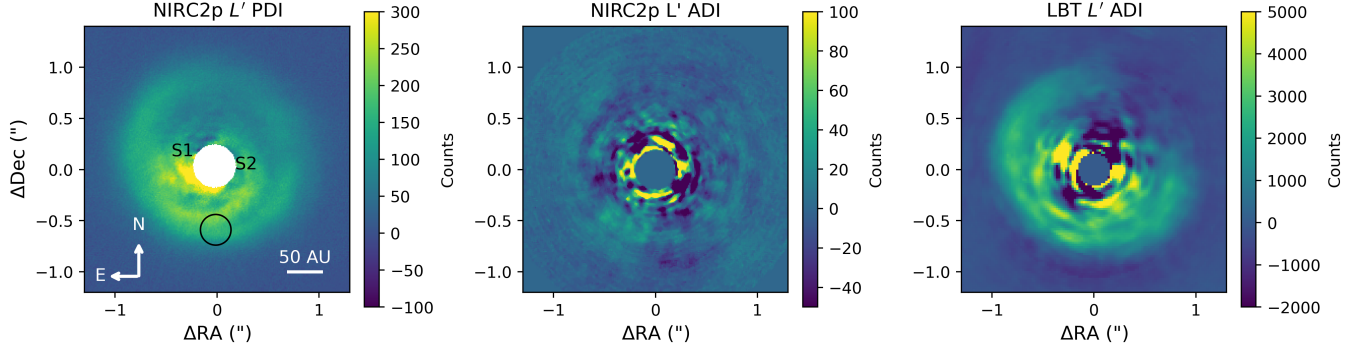


Figure 3. (Left) AB Aur in L' Stokes Q_ϕ with spiral arms S1 and S2 annotated, along with a circle centered on the coordinates of the candidate protoplanet from Currie et al. (2025). (Middle) AB Aur in L' total intensity derived from the NIRC2 Polarimetry data with ADI. (Right) For comparison: AB Aur seen in total intensity L' with KLIP-RDI from Betti et al. (2022). It is clear that L' polarimetry is able to recover more detail of the disk's spiral structure and extended emission than was possible with traditional PSF subtraction methods on total intensity images.

Pol is unique, as Keck II is the largest telescope (10 m) on which infrared polarimetry capabilities are available, and the only that can be combined with a vortex coronagraph. The new mode has been rigorously tested in both daytime and on-sky commissioning observations, and is now available to the community. Rigorous quantitative calibration of the system's instrumental polarization is underway (Zhang et al. in preparation), and a data processing pipeline is in development to aid potential observers (Lewis et al. in preparation). First on-sky results with NIRC2-Pol do not detect the candidate protoplanet AB Aur b—nor clear evidence of a dust clump at that location—but do provide the first L' polarimetry of AB Aur's disk and the clearest L' imaging of its spiral features to date, illustrating the potential of this new mode.

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Facilities: Keck II (NIRC2)

Software: NumPy (van der Walt et al. 2011), IPython (Perez & Granger 2007), Jupyter Notebooks (Kluyver et al. 2016), Matplotlib (Hunter 2007), Astropy (Astropy Collaboration et al. 2013; Price-Whelan et al. 2018), SciPy (Jones et al. 2001) ADI.jl (Lucas & Bottom 2020)

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